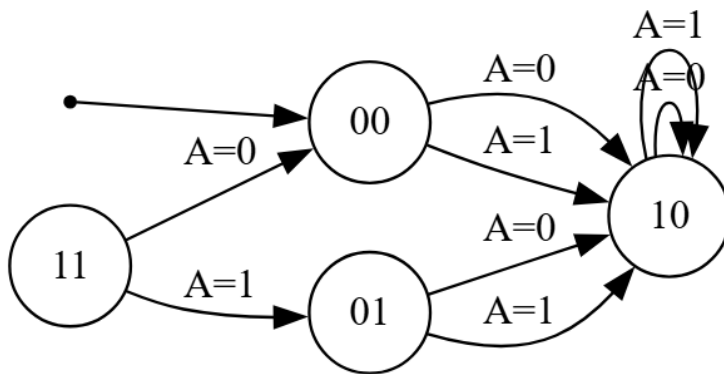
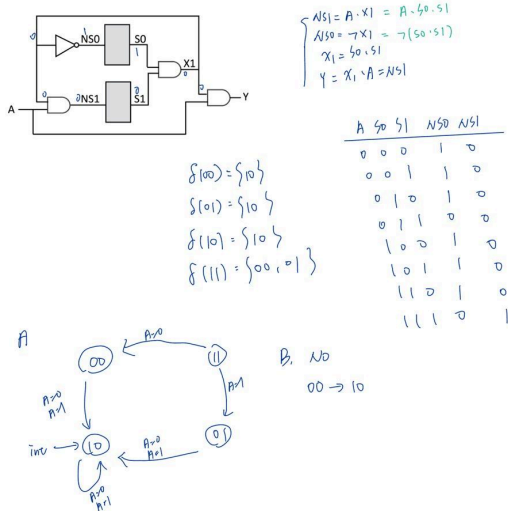


ECE 622 Lab 1

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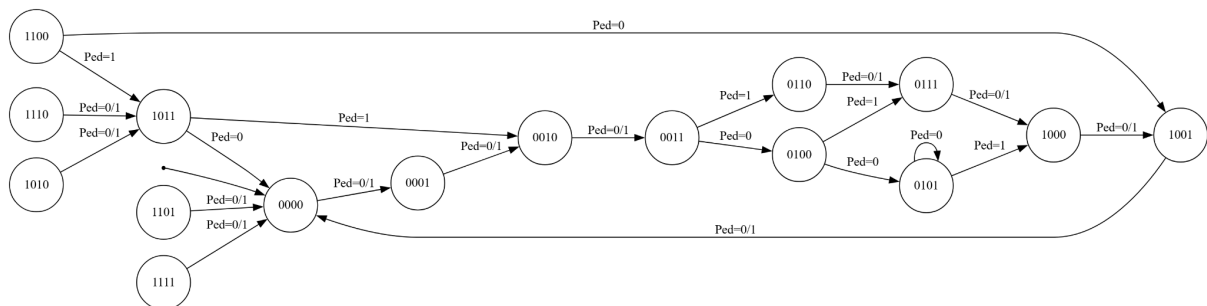
A.1

ex1: use the following method to draw ex1 transition state graph



stoplight1: write a testbench for stoplight module, used to test all state transitions and display the results based on changes in ped

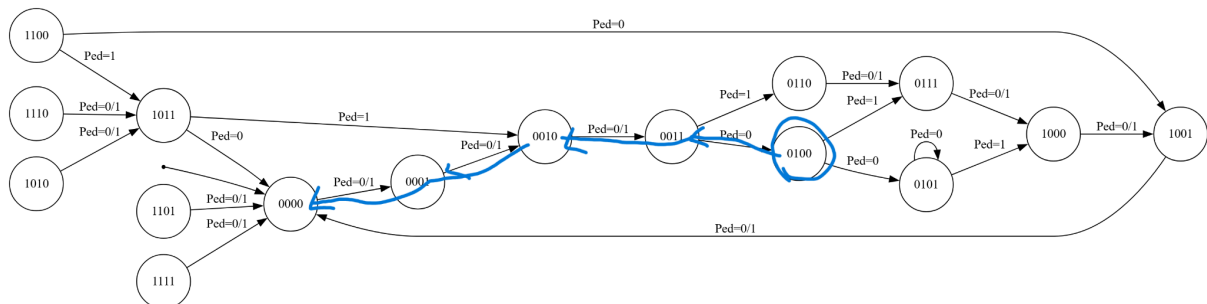
S=0000	Ped=0 -> 0001	Ped=1 -> 0001
S=0001	Ped=0 -> 0010	Ped=1 -> 0010
S=0010	Ped=0 -> 0011	Ped=1 -> 0011
S=0011	Ped=0 -> 0100	Ped=1 -> 0110
S=0100	Ped=0 -> 0101	Ped=1 -> 0111
S=0101	Ped=0 -> 0101	Ped=1 -> 1000
S=0110	Ped=0 -> 0111	Ped=1 -> 0111
S=0111	Ped=0 -> 1000	Ped=1 -> 1000
S=1000	Ped=0 -> 1001	Ped=1 -> 1001
S=1001	Ped=0 -> 0000	Ped=1 -> 0000
S=1010	Ped=0 -> 1011	Ped=1 -> 1011
S=1011	Ped=0 -> 0000	Ped=1 -> 0010
S=1100	Ped=0 -> 1001	Ped=1 -> 1011
S=1101	Ped=0 -> 0000	Ped=1 -> 0000
S=1110	Ped=0 -> 1011	Ped=1 -> 1011
S=1111	Ped=0 -> 0000	Ped=1 -> 0000



A.2

ex1: according to transition state graph 00 will never reach target state 11

stoplight1: 0000(ped=0/1)->0001(ped=0/1)->0010(ped=0/1)->0011(ped=0)->0100



A.3

ex1:

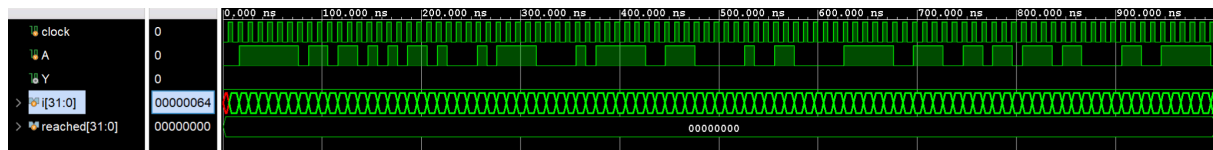
A	B
1	1
2	1
3	1
4	1
5	1
6	1
7	1
8	1
9	1
10	1
11	1
12	1
13	1
14	1
15	1
16	1
17	1
18	1
19	1
20	1

stoplight1:

1	1
2	2
3	3
4	5
5	7
6	8
7	9
8	9
9	9
10	9
11	9
12	9
13	9
14	9
15	9
16	9
17	9
18	9
19	9
20	9

A.4

ex1: initial state will never reach target state

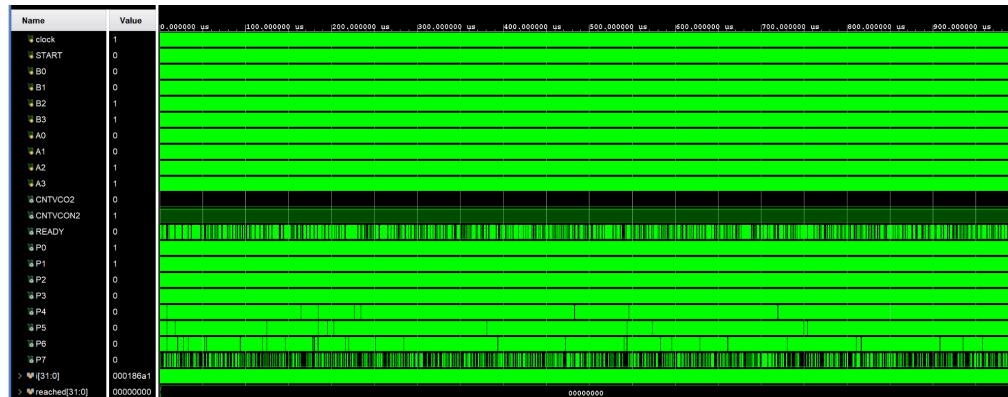


```

ex1: Target state 11 NOT reached within 100000 cycles (TIMED OUT)
$finish called at time : 1000006 ns : File "C:/Users/maxf2/Desktop/embedded system/lab1_files/tb_ex1.v" Line 51

```

ex2: initial state will never reach target state

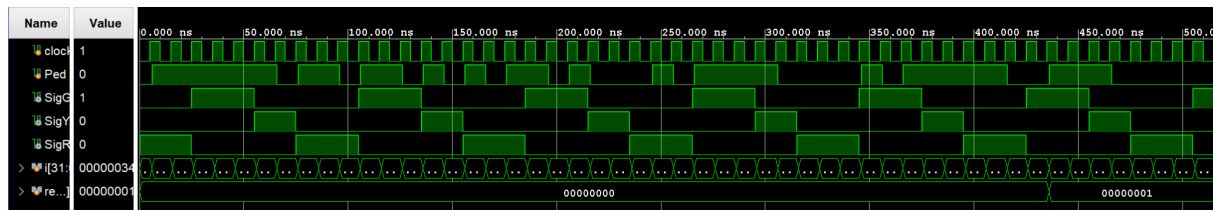


```

# run 1000ns
INFO: [USF-XSim-96] XSim completed. Design snapshot 'tb_ex2_behav' loaded.
INFO: [USF-XSim-97] XSim simulation ran for 1000ns
launch_simulation: Time (s): cpu = 00:00:03 ; elapsed = 00:00:10 . Memory (MB): peak = 2!
run -all
ex2: Target state 11111111111111 NOT reached within 100000 cycles (TIMED OUT)

```

stoplight1:

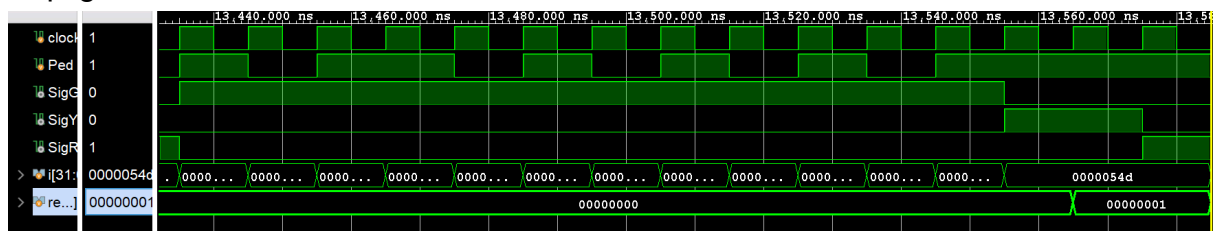


```

INFO: [USF-XSim-97] XSim simulation ran for 1000ns
launch_simulation: Time (s): cpu = 00:00:03 ; elapsed = 00:00:10 . Memory (MB): peak = 2562.023 ; gain = 0.000
ERROR: [Common 17-180] Spawn failed: No error
run -all
stoplight1: Target state 0100 REACHED at cycle 52
$finish called at time : 516 ns : File "C:/Users/maxf2/Desktop/embedded system/lab1_files/tb_soptlight1.v" Line 39

```

stopligh2:



```

INFO: [USF-XSim-96] XSim completed. Design snapshot 'tb_stoplight2_behav' loaded.
INFO: [USF-XSim-97] XSim simulation ran for 1000ns
launch_simulation: Time (s): cpu = 00:00:12 ; elapsed = 00:00:11 . Memory (MB): peak = 2658.184 ; gain = 0.000
run -all
stoplight2: Target state 01000 REACHED at cycle 1357
$finish called at time : 13585 ns : File "C:/Users/maxf2/Desktop/embedded system/lab1 files/tb stoplight2.v" Line 41

```

stoplight1 testbench

```
1 `timescale 1ns/1ps
2
3 module tb_stoplight1;
4     reg clock, Ped;
5     wire SigG, SigY, SigR;
6     integer i;
7     integer reached;
8
9     stoplight1 spotlight1 (
10         .clock(clock),
11         .Ped(Ped),
12         .SigG(SigG),
13         .SigY(SigY),
14         .SigR(SigR)
15     );
16
17     initial clock = 0;
18     always #5 clock = ~clock;
19
20     initial begin
21         reached = 0;
22         Ped = 0;
23         spotlight1.S3 = 0;
24         spotlight1.S2 = 0;
25         spotlight1.S1 = 0;
26         spotlight1.S0 = 0;
27
28         for (i = 1; i <= 100000; i = i + 1) begin
29             Ped = $random % 2;
30             @(posedge clock);
31
32             // Target:0100
33             if (spotlight1.S3 == 0 && spotlight1.S2 == 1 && spotlight1.S1 == 0 && spotlight1.S0 == 0) begin
34                 $display("stoplight1: Target state 0100 REACHED at cycle %0d", i);
35                 reached = 1;
36                 @(posedge clock);
37                 @(posedge clock);
38                 $finish;
39             end
40         end
41
42         if (reached == 0)
43             $display("stoplight1: Target state 0100 NOT reached within 100000 cycles (TIMED OUT)");
44
45         $finish;
46     end
47 endmodule
```

A.5

The result for stoplight1 is consistent with the result from A.2. We know that stoplight1 will reach the target state 0100 in A.2 and A.4 simulation shows the same result.

B.1

State '11' is not reachable from the initial state 00 in 2 transitions. This is consistent with the state transition graph from A.1, since clearly the FSM, if initialized at 00, gets stuck at 01.

```

jdnotlurking@14art-he:~/622modeling/lab1$ ./lab1_parser ./verilog_src/ex1.v 11 2 00 results_ex1 minisat
WARNING: for repeatability, setting FPU to use double precision
=====
[ Problem Statistics ]=====
|
| Number of variables:      18
| Number of clauses:       30
| Parse time:              0.00 s
| Simplification time:     0.00 s
|
|=====
Solved by simplification
restarts      : 0
conflicts    : 0          (-nan /sec)
decisions    : 0          (-nan % random) (-nan /sec)
propagations : 10         (inf /sec)
conflict literals : 0      (-nan % deleted)
Memory used  : 22.00 MB
CPU time     : 0 s

```

Repeating similar analysis for other states, and mentioning the final result:

State 00: UNSAT. The graph *has* to move, it can not stall/self-loop at 00.

```

jdnotlurking@14art-he:~/622modeling/lab1$ ./lab1_parser ./verilog_src/ex1.v 00 2 00 results_ex1 minisat
WARNING: for repeatability, setting FPU to use double precision
=====
[ Problem Statistics ]=====
|
| Number of variables:      18
| Number of clauses:       30
| Parse time:              0.00 s
| Simplification time:     0.00 s
|
|=====
Solved by simplification
restarts      : 0
conflicts    : 0          (0 /sec)
decisions    : 0          (-nan % random) (0 /sec)
propagations : 10         (5405 /sec)
conflict literals : 0      (-nan % deleted)
Memory used  : 22.00 MB
CPU time     : 0.00185 s

UNSATISFIABLE
WARNING: minisat returned non-zero exit code 5120
Wrote results_ex1/out.dimacs and results_ex1/out.nodes
State regs inferred (target bit order assumed to match this order): S1 S0
jdnotlurking@14art-he:~/622modeling/lab1$

```

State 01: SAT. Reachable in 2 states. Any value of 'k' {k >= 1} unrolling will be SAT due to the self loop.

```

jdnotlurking@14art-he:~/622modeling/lab1$ ./lab1_parser ./verilog_src/ex1.v 01 2 00 results_ex1 minisat
WARNING: for repeatability, setting FPU to use double precision
=====
[ Problem Statistics ]=====
|
| Number of variables:      18
| Number of clauses:       30
| Parse time:              0.00 s
| Eliminated clauses:      0.00 Mb
| Simplification time:     0.00 s
|
|=====
===== [ Search Statistics ] =====
| Conflicts | ORIGINAL | LEARNED | Progress |
|           | Vars  Clauses Literals | Limit  Clauses Lit/Cl |
|=====|=====|=====|=====|
restarts      : 1
conflicts    : 0          (0 /sec)
decisions    : 1          (0.00 % random) (717 /sec)
propagations : 14         (10036 /sec)
conflict literals : 0      (-nan % deleted)
Memory used  : 22.00 MB
CPU time     : 0.001395 s

SATISFIABLE
WARNING: minisat returned non-zero exit code 2560
Wrote results_ex1/out.dimacs and results_ex1/out.nodes
State regs inferred (target bit order assumed to match this order): S1 S0
jdnotlurking@14art-he:~/622modeling/lab1$

```

State 10: UNSAT. 10 is not reachable from 00 .

```
jdnotlurking@14art-he:~/622modeling/lab1$ ./lab1_parser ./verilog_src/ex1.v 10 2 00 results_ex1 minisat
WARNING: for repeatability, setting FPU to use double precision
===== [ Problem Statistics ] =====
|
| Number of variables:          18
| Number of clauses:           30
| Parse time:                  0.00 s
| Simplification time:         0.00 s
|
|=====
Solved by simplification
restarts      : 0
conflicts    : 0                (0 /sec)
decisions    : 0                (-nan % random) (0 /sec)
propagations : 10               (4850 /sec)
conflict literals : 0          (-nan % deleted)
Memory used  : 22.00 MB
CPU time     : 0.002062 s

UNSATISFIABLE
WARNING: minisat returned non-zero exit code 5120
Wrote results_ex1/out.dimacs and results_ex1/out.nodes
State regs inferred (target bit order assumed to match this order): S1 S0
```

B.2

1. ex1

```
jdnotlurking@14art-he:~/622modeling/lab1$ ./check_reachability.sh ./verilog_src/ex1.v 11 15 00 logs_reachability minisat
=== RUN @ 20260226_212011 ===
Verilog : ./verilog_src/ex1.v
Init     : 00
Target   : 11
Unroll k : 15
Outdir   : logs_reachability
Minisat  : minisat
Flags    : print=false state_only=false summary=false

WARNING: for repeatability, setting FPU to use double precision
===== [ Problem Statistics ] =====
|
| Number of variables:          122
| Number of clauses:           225
| Parse time:                  0.00 s
| Simplification time:         0.00 s
|
|=====
Solved by simplification
restarts      : 0
conflicts    : 0                (-nan /sec)
decisions    : 0                (-nan % random) (-nan /sec)
propagations : 10               (inf /sec)
conflict literals : 0          (-nan % deleted)
Memory used  : 22.00 MB
CPU time     : 0 s

UNSATISFIABLE
WARNING: minisat returned non-zero exit code 5120
Wrote logs_reachability/out.dimacs and logs_reachability/out.nodes
State regs inferred (target bit order assumed to match this order): S1 S0
=== Full SAT Output ===
UNSAT

=== UNSAT: Skipping nodes/model printing ===

=== END RUN ===

Saved log: logs_reachability/logs/ex1_init00_target11_k15_20260226_212011.txt
jdnotlurking@14art-he:~/622modeling/lab1$
```

2. ex2


```

jdoetlurking@14art-he: /622modeling/lab1$ ./check_reachability.sh ./verilog_src/ex2.v 11111111111111 15 00000000000000 logs_reachability minisat
=== RUN @ 20260226_211904 ===
Verilog : ./verilog_src/ex2.v
Init : 000000000000000000000000
Target : 1111111111111111
Unroll k : 15
Outdir : logs_reachability
Minisat : minisat
Flags : print=false state_only=false summary=false

WARNING: minisat returned non-zero exit code 5120
Wrote logs_reachability/out.dinacs and logs_reachability/out.nodes
State regs inferred (target bit order assumed to match this order): S14 S13 S12 S11 S10 S9 S8 S7 S6 S5 S4 S3 S2 S1 S0
WARNING: for repeatability, setting FPU to use double precision
===== [ Problem Statistics ] =====
|
| Number of variables:      3540
| Number of clauses:       8385
| Parse time:              0.00 s
| Eliminated clauses:      0.09 Mb
| Simplification time:     0.01 s
|
|===== [ Search Statistics ] =====
| Conflicts | Vars | ORIGINAL | LEARNNT | Progress |
| Clauses | Literals | Limit | Clauses | Lit/cl |
|=====|=====|=====|=====|=====|
restarts      : 1
conflicts     : 2          (182 /sec)
decisions     : 3          (0.00 % random) (272 /sec)
propagations  : 584        (53019 /sec)
conflict literals : 1      (0.00 % deleted)
Memory used   : 23.00 MB
CPU time      : 0.011015 s

UNSATISFIABLE
=== Full SAT Output ===
UNSAT

=== UNSAT: Skipping nodes/model printing ===

=== END RUN ===

Saved log: logs_reachability/logs/ex2_init0000000000000000_target11111111111111_k15_20260226_211904.txt
jdoetlurking@14art-he: /622modeling/lab1$

```

3. ex3

```

jdoetlurking@14art-he: /622modeling/lab1$ ./check_reachability.sh ./verilog_src/ex3.v 000000000101101100000110000001 15 000000000000000000000000 logs_reachability minisat -print-state-only
=== RUN @ 20260226_211600 ===
Verilog : ./verilog_src/ex3.v
Init : 00000000000000000000000000000000
Target : 000000000101101100000110000001
Unroll k : 15
Outdir : logs_reachability
Minisat : minisat
Flags : print=false state_only=true summary=false

WARNING: for repeatability, setting FPU to use double precision
===== [ Problem Statistics ] =====
|
| Number of variables:      9874
| Number of clauses:       22785
| Parse time:              0.00 s
| Eliminated clauses:      0.18 Mb
| Simplification time:     0.01 s
|
|===== [ Search Statistics ] =====
| Conflicts | Vars | ORIGINAL | LEARNNT | Progress |
| Clauses | Literals | Limit | Clauses | Lit/cl |
|=====|=====|=====|=====|=====|
restarts      : 1
conflicts     : 37          (2156 /sec)
decisions     : 263        (0.00 % random) (15325 /sec)
propagations  : 3617       (210760 /sec)
conflict literals : 145    (6.45 % deleted)
Memory used   : 25.00 MB
CPU time      : 0.017161 s

SATISFIABLE
WARNING: minisat returned non-zero exit code 2500
Wrote logs_reachability/out.dinacs and logs_reachability/out.nodes
State regs inferred (target bit order assumed to match this order): S28 S27 S26 S25 S24 S23 S22 S21 S20 S19 S18 S17 S16 S15 S14 S13 S12 S11 S10 S9 S8 S7 S6 S5 S4 S3 S2 S1 S0
=== Full SAT Output ===
SAT
-1 -2 -3 -4 -5 -6 -7 -8 -9 -10 -11 -12 -13 -14 -15 -16 -17 -18 -19 -20 -21 -22 -23 -24 -25 -26 -27 -28 -29 -30 31 32 33 34 -35 -36 -37 -38 -39 -40 41 -42 -43 -44 45 -46 -47 -48 -49 -50 51 -52 53 54
7 58 -59 60 61 62 63 64 65 66 -67 -68 69 -70 -71 -72 -73 74 -75 -76 -77 78 79 -80 -81 -82 83 -84 85 86 -87 88 -89 -90 -91 -92 -93 94 95 96 97 -98 -99 100 -101 -102 -103 104 -105 106 -107 108 109
-112 113 -114 -115 116 117 -118 -119 120 -121 122 123 124 -125 -126 -127 128 129 -130 -131 -132 133 -134 -135 -136 137 138 -139 -140 141 142 143 144 145 -146 -147 148 -149 150 151 -152 -153 154 15
-157 -158 -159 160 -161 -162 163 -164 -165 -166 -167 -168 -169 170 171 -172 173 -174 175 -176 177 -178 -179 180 -181 -182 183 -184 185 -186 -187 188 -189 189 190 191 192 -193 194 195 -196 -197 198 199 200

```

4. ex4


```

jdnofurking@i4art-he:~/622modeling/lab1$ ./check_reachability.sh ./verilog_src/ex4.v 00000000000000000000000000000000 15 000000000000000000000000000000 logs_reachability minisat
=== RUN @ 20260226_212117 ===
Verilog : ./verilog_src/ex4.v
Init : 00000000000000000000000000000000
Target : 00000000000000000000000000000000
Unroll k : 15
Outdir : logs_reachability
Minisat : minisat
Flags : print=false state_only=false summary=false

WARNING: minisat returned non-zero exit code 5120
WARNING: for repeatability, setting FPU to use double precision
===== [ Problem Statistics ] =====
|
| Number of variables:      7577
| Number of clauses:       10210
| Parse time:              0.00 s
| Eliminated clauses:      0.05 Mb
| Simplification time:     0.00 s
|
=====
Solved by simplification
restarts      : 0
conflicts    : 0                (0 /sec)
decisions    : 0                (-nan % random) (0 /sec)
propagations : 4956             (514802 /sec)
conflict literals : 0          (-nan % deleted)
Memory used  : 24.00 MB
CPU time     : 0.009627 s

UNSATISFIABLE
Wrote logs_reachability/out.dimacs and logs_reachability/out.nodes
State regs inferred (target bit order assumed to match this order): S31 S30 S29 S28 S27 S26 S25 S24 S23 S22 S21 S20 S19 S18 S17 S16 S15 S14 S13 S12 S11 S10 S9 S8 S7 S6 S5 S4 S3 S2 S1 S0
=== Full SAT Output ===
UNSAT

=== UNSAT: Skipping nodes/model printing ===

=== END RUN ===

Saved log: logs_reachability/logs/ex4_init00000000000000000000000000000000_target00000000000000000000000000000000_k15_20260226_212117.txt
jdnofurking@i4art-he:~/622modeling/lab1$

```

5. ex5

```

jdnofurking@i4art-he:~/622modeling/lab1$ ./check_reachability.sh ./verilog_src/ex5.v 000110111 15 000000000 logs_reachability minisat
=== RUN @ 20260226_212315 ===
Verilog : ./verilog_src/ex5.v
Init : 000000000
Target : 000110111
Unroll k : 15
Outdir : logs_reachability
Minisat : minisat
Flags : print=false state_only=false summary=false

WARNING: for repeatability, setting FPU to use double precision
===== [ Problem Statistics ] =====
|
| Number of variables:      3444
| Number of clauses:       8205
| Parse time:              0.00 s
| Simplification time:     0.00 s
|
=====
Solved by simplification
restarts      : 0
conflicts    : 0                (0 /sec)
decisions    : 0                (-nan % random) (0 /sec)
propagations : 245             (63031 /sec)
conflict literals : 0          (-nan % deleted)
Memory used  : 23.00 MB
CPU time     : 0.003887 s

UNSATISFIABLE
WARNING: minisat returned non-zero exit code 5120
Wrote logs_reachability/out.dimacs and logs_reachability/out.nodes
State regs inferred (target bit order assumed to match this order): S8 S7 S6 S5 S4 S3 S2 S1 S0
=== Full SAT Output ===
UNSAT

=== UNSAT: Skipping nodes/model printing ===

=== END RUN ===

Saved log: logs_reachability/logs/ex5_init000000000_target000110111_k15_20260226_212315.txt
jdnofurking@i4art-he:~/622modeling/lab1$

```

6. stoplight1

```

jdn0turlking@i4art-he:~/622modeling/lab1$ ./check_reachability.sh ./verilog_src/stoplight1.v 0100 15 0000 logs_reachability minisat
=== RUN @ 20260226_212424 ===
Verilog : ./verilog_src/stoplight1.v
Init    : 0000
Target  : 0100
Unroll k: 15
Outdir  : logs_reachability
Minisat : minisat
Flags   : print=false state_only=false summary=false

WARNING: for repeatability, setting FPU to use double precision
===== [ Problem Statistics ] =====
|
| Number of variables:      934
| Number of clauses:       2310
| Parse time:              0.00 s
| Eliminated clauses:      0.02 Mb
| Simplification time:     0.00 s
|
===== [ Search Statistics ] =====
| Conflicts |      Vars | ORIGINAL |      LEARNT | Progress |
|   Clauses |   Literals |   Clauses |   Lit/Cl   |          |
=====
restarts      : 1
conflicts     : 11          (3204 /sec)
decisions     : 32          (0.00 % random) (9321 /sec)
propagations  : 731        (212933 /sec)
conflict literals : 24      (4.00 % deleted)
Memory used   : 22.00 MB
CPU time      : 0.003433 s

SATISFIABLE
WARNING: minisat returned non-zero exit code 2560
Wrote logs_reachability/out.dimacs and logs_reachability/out.nodes
State regs inferred (target bit order assumed to match this order): S3 S2 S1 S0
=== Full SAT Output ===
SAT
-1 -2 -3 -4 -5 -6 -7 -8 -9 -10 -11 -12 -13 -14 -15 -16 -17 -18 -19 -20 -21 -22 -23 -24 -25 -26 -27 -28 -29 -30 -31 -32 -33 -34 -35 -36 -37 -38 -39 -40 -
61 62 -63 -64 -65 66 67 -68 69 -70 71 -72 73 -74 -75 76 77 -78 -79 80 81 82 -83 84 85 -86 -87 -88 89 90 91 92 -93 -94 -95 96 97 -98 -
-115 116 117 -118 119 120 121 122 123 -124 -125 -126 127 -128 129 -130 131 -132 -133 -134 135 -136 -137 138 139 140 -141 142 143 144 -
160 161 -162 -163 164 -165 166 167 -168 -169 170 -171 -172 173 174 -175 -176 -177 178 -179 -180 181 -182 -183 -184 -185 186 -187 -188

```

7. stoplight2

```

jdn0turlking@i4art-he:~/622modeling/lab1$ ./check_reachability.sh ./verilog_src/stoplight2.v 01000 15 00000 logs_reachability minisat
=== RUN @ 20260226_212522 ===
Verilog : ./verilog_src/stoplight2.v
Init    : 00000
Target  : 01000
Unroll k: 15
Outdir  : logs_reachability
Minisat : minisat
Flags   : print=false state_only=false summary=false

WARNING: for repeatability, setting FPU to use double precision
===== [ Problem Statistics ] =====
|
| Number of variables:      1715
| Number of clauses:       4455
| Parse time:              0.00 s
| Eliminated clauses:      0.03 Mb
| Simplification time:     0.00 s
|
===== [ Search Statistics ] =====
| Conflicts |      Vars | ORIGINAL |      LEARNT | Progress |
|   Clauses |   Literals |   Clauses |   Lit/Cl   |          |
=====
restarts      : 1
conflicts     : 46          (9662 /sec)
decisions     : 70          (0.00 % random) (14703 /sec)
propagations  : 2694        (565848 /sec)
conflict literals : 252      (23.87 % deleted)
Memory used   : 22.00 MB
CPU time      : 0.004761 s

SATISFIABLE
WARNING: minisat returned non-zero exit code 2560
Wrote logs_reachability/out.dimacs and logs_reachability/out.nodes
State regs inferred (target bit order assumed to match this order): S4 S3 S2 S1 S0
=== Full SAT Output ===
SAT
-1 -2 -3 -4 -5 -6 -7 8 9 10 11 -12 -13 -14 -15 -16 -17 -18 -19 -20 -21 -22 -23 -24 -25 -26 -27 -28 -29 -30 -31 -32 -33 -34 -35 -36 -37 -38 -39 -40 -41 -
62 -63 -64 65 66 67 68 -69 -70 -71 72 -73 74 -75 -76 77 78 79 -80 -81 -82 -83 84 -85 -86 87 88 89 90 -91 -92 93 -94 -95 96 97 -98 -99 -
-116 -117 -118 119 120 -121 -122 123 124 125 -126 -127 128 129 -130 -131 -132 133 134 -135 136 137 138 139 140 -141 -142 -143 144 145 -
161 162 -163 -164 165 -166 167 168 -169 -170 171 -172 173 -174 -175 176 -177 -178 179 -180 181 182 -183 -184 -185 186 -187 188 -189 -190

```

>TODO (self): add picosat results

B.3

For stoplight1.v; target state 0100 is reachable for all the values of i above (and including) 12.

Answer: $i = \{12, 13, 14 \dots 31, 32\}$

B.4

For stoplight2.v; target state 0100 is reachable for all the values of i above (and including) 14.

Answer: $i = \{14, 15 \dots 31, 32\}$

B.5

Approach to find $\text{delta}(S_0)$ with $k=17$:

- The SAT instance encodes the unrolled transition relation for $k=17$, starting from the fixed initial state at time 0.
- A candidate state “s” is enforced by adding unit clauses that constrain the state variables at time 17 to equal s.
- If the SAT solver returns SAT, there exists an input sequence of length 17 taking the machine from initial $\rightarrow$ s, hence s is reachable at transition
- Repeat this for all $2^5 = 32$ possible states; collect the ones that are SAT.

Following the steps as mentioned, we got the following 19 states to be reachable at $k=17$ th unrolling of stoplight2.v:

{00000, 00001, 00010, 00011, 00100, 00101, 00110, 00111, 01000, 01001, 10000, 10001, 10010, 10011, 10100, 10101, 10110, 10111, 11000}

C

First, we parse verilog, read the file and extract primary inputs, registers. Second, we build evaluator

ex1:

1		1
2		1
3		1
4		1
5		1
6		1
7		1
8		1
9		1
10		1
11		1
12		1
13		1
14		1
15		1
16		1
17		1
18		1
19		1
20		1

ex5:

1		8
2		14
3		18
4		22
5		26
6		28
7		28
8		28
9		28
10		28
11		28
12		28
13		28
14		28
15		28
16		28
17		28
18		28
19		28
20		28